

Rajiv Gandhi Institute of Petroleum Technology (RGIPT)
Department of Computer Science & Engineering
CS441 – Generative Artificial Intelligence Laboratory
Laboratory Assignment–1

Objective

Explore the Hugging Face ecosystem, implement pretrained AI pipelines using the Transformers library, investigate model repositories and Hugging Face Spaces, and analyze the behaviour of pretrained foundation models.

Problem Statement

Hugging Face provides an ecosystem for discovering, evaluating and deploying pretrained AI models. Through investigation and implementation, study how pretrained models are organized, executed and documented, and evaluate their practical utility.

Tasks to be performed

1. Using the provided practice notebook, generate images using prompts: (a) 'A futuristic smart city' and (b) 'A futuristic smart city with autonomous vehicles, glass skyscrapers, cinematic lighting, ultra realistic, 8K'. Experimentally evaluate how prompt refinement affects semantic alignment and visual quality, and suggest an improvement in the prompt.
2. Provide one practical use case with Model Hub, Dataset Hub, Spaces, Organizations and Collections in hugging face ecosystem and summarize its purpose.
3. Explore the Hugging Face Model Hub (<https://huggingface.co/models>) and identify THREE popular text-to-image conversion models and analyze its Model Cards. Report the available description along with limitations and repository files.
4. Implement Hugging Face pipelines for Sentiment Analysis, Text Generation, Summarization, Translation, Question Answering, Image Classification. For each implementation include: (i) source code, (ii) input, (iii) output, (iv) screenshot, and (v) a brief technical observation.
5. Select one public Hugging Face Space. Identify the SDK (Gradio/Streamlit/Docker), describe its functionality, execute the application, include screenshots, and explain how Spaces differ from Models and Datasets.

Deliverables

- Submit a report (jupyter notebook) containing the implementation for all the above 5 tasks.

Expected Learning Outcomes

- Navigate the Hugging Face ecosystem.
- Interpret technical information from Model Cards.
- Perform inference using a pretrained model.
- Critically analyze experimental observations with hugging face ecosystem